

Intelligent Elevator System

Natural Language Interface & Predictive Traffic Management

A smart, accessible, and touchless building elevator ecosystem
powered by HCI · NLP · Machine Learning

HCI

NLP

ML / AI

Accessibility

PRESENTER

Mya Mjechal

COURSE

AIEF Class

Dr. Ye Kyaw Thu

FL 5

FL 4

FL 3

FL 2

FL 1

5 User Pain Points



User POV

Physical Interaction

Pressing shared buttons is unhygienic (post-COVID), unreachable for wheelchair users, and hard for those carrying heavy items



User POV

Waiting Time

Simple algorithms ignore daily building patterns e.g. morning rush, lunch visits, evening exits which causing long unpredictable waits



User POV

No Urgency Signal

A person rushing to a meeting is treated the same as someone going to lunch. No way to request priority or say 'I'm in a hurry'



User POV

Accessibility

Visually impaired users can't identify arriving elevators. Deaf users miss closing-door warnings. Audio cues are often missing or unclear



User POV

No Information

No data on wait time, which elevator is coming, or how crowded it is creating anxiety especially during peak hours

Real-Time Information Display

Give users full visibility of wait time, which elevator is coming, and how crowded it is.

- Smart panels show estimated wait time per direction
- Which specific elevator (A, B, C) is arriving
- Crowding level: green = space · yellow = moderate · red = crowded
- Mobile push: "Your elevator arrives in 30 seconds"

LOBBY DISPLAY PANEL

Elevator A

FL8
FL7
FL6
FL5
FL4
FL3
FL2
FL1

ETA: 45s

Crowded

Elevator B

FL8
FL7
FL6
FL5
FL4
FL3
FL2
FL1

ETA: 12s

Available

Elevator C

FL8
FL7
FL6
FL5
FL4
FL3
FL2
FL1

ETA: 1m5s

Moderate

 Available Moderate Crowded

Crowd Detection & Predictive Dispatch

Computer vision counts waiting users floor-by-floor and dispatches elevators before queues form.

01

Privacy-preserving cameras

Count people on each floor and inside elevators, no identifiable images stored

02

Idle elevator re-dispatch

System redirects an idle elevator from a quiet floor to serve the crowded floor.

03

Less-crowded routing

Suggests the least crowded elevator to users via display panels and mobile app

04

Prevents overcrowding

Monitors elevator capacity and gates entry automatically

Floor 5 (Offices)



1 waiting



Floor 4 (Offices)



2 waiting



Floor 3 (Meeting Rms)



5 waiting



Floor 2 (Cafeteria)



4 waiting



Floor 1 (Lobby)



8 waiting



Conversational Elevator Interface

Speak or type naturally — the elevator understands intent, floor, and urgency.

- "Where is the cafeteria?" → elevator called to floor 2
- "Take me to Dr. Smith's office" → system resolves the floor
- "I'm in a hurry" → priority dispatch requested
- Accessibility: voice + audio cues for visually impaired

CONVERSATION DEMO

Where is the cafeteria?

Cafeteria is on Floor 2. Should I call an elevator going down?

Yes, and I'm in a hurry.

Priority request sent! Elevator B arriving in 12 seconds — Floor 5 door opening now.

ELEVATOR B — NOW DISPATCHED

1

2

3

4

5

Current: FL5 Target: FL2

What This System Delivers

Wait Time Reduction

Predictive dispatch learns daily building patterns — morning rush, lunch, evening — and pre-positions elevators

Touchless Operation

Full voice + NLP control eliminates need to touch any shared button surface

Full Accessibility

Audio announcements, voice commands, and visual floor indicators serve visually impaired and deaf users

User Satisfaction

Real-time display panels and urgency signaling eliminate uncertainty and frustration

Energy Efficiency

Smart dispatch avoids sending empty or near-empty elevators — reduces idle trips

Emergency Response

System supports priority floor lockdown and emergency dispatch integration